The Kalman-Bucy filter

Suppose that $B(x) = Bx$ and $H(x) = Hx$ are linear, $C(x) = C$ constant and $X_0 \sim N(m_0, P_0)$. Then

$$d\begin{bmatrix} X_t \\ Y_t \end{bmatrix} = \begin{bmatrix} B & 0 \\ H & 0 \end{bmatrix}\begin{bmatrix} X_t \\ Y_t \end{bmatrix} dt + \begin{bmatrix} C & 0 \\ 0 & \Gamma \end{bmatrix}\begin{bmatrix} dW_t \\ dV_t \end{bmatrix}$$

$\begin{bmatrix} X_0 \\ Y_0 \end{bmatrix} \sim N\left(\begin{bmatrix} m_0 \\ 0 \end{bmatrix}, \begin{bmatrix} P_0 & 0 \\ 0 & 0 \end{bmatrix}\right)$, is a Gaussian process, i.e., its distribution on $C([0,\infty); \mathbb{R}^d \times \mathbb{R}^p)$ is a Gaussian measure.
In particular, $\eta_t = N(m_t, P_t)$ is again a normal distribution (without proof) and m_t, P_t can be computed, using the <u>Kalman-Bucy filtering equations:</u>

(2a) $dm_t = Bm_t dt + P_t H^T R^{-1}(dY_t - Hm_t dt)$

(2b) $\dot{P}_t = \frac{d}{dt}P_t = BP_t + P_t B^T + CC^T - P_t H^T R^{-1} H P_t$

<u>Note:</u> (2b) is a matrix Riccati differential equation taking values in the set of positive definite symmetric matrices. The solution is independent of the observation (Y_t) and can thus be computed a priori.

The solution m_t of the SDE (2a) is $E(X_t \mid Y_{0:t})$, i.e., the conditional mean. The conditional variance P_t yields the L^2-mean square estimation error since

$$\mathbb{E}\left(\|X_t - m_t\|^2\right) = \mathbb{E}\left(\mathbb{E}\left(\|X_t - m_t\|^2 \mid Y_{0:t}\right)\right) = \operatorname{tr}(P_t).$$

The following equations (2a,b) can be derived from explicit solutions of the K-S equation as follows:

Take $f(x) = e^{iux}$, hence $\eta_t(f) = \int e^{i\langle u,x\rangle}\eta_t(dx) = e^{i\langle m_t,u\rangle - \frac{1}{2}\langle u,P_t u\rangle}$ is the characteristic function on η_t. Assuming $\eta_t = N(m_t, P_t)$ is Gaussian (without proof here) we can compute:

$$d\eta_t(e^{i\langle u,x\rangle}) = \eta_t(Le^{i\langle u,x\rangle}) + \mathrm{cov}_{\eta_t}(e^{i\langle u,x\rangle}, Hx)R^{-1}(dY_t - \eta_t(Hx)dt)$$

$$= \left(-\tfrac{1}{2}\langle CC^T u,u\rangle + i\langle B^T u, m_t + iP_t u\rangle\right)\eta_t(e^{i\langle u,x\rangle})dt$$

$$\boxed{\begin{aligned} Le^{i\langle u,x\rangle} &= \left(-\tfrac{1}{2}\langle CC^T u,u\rangle + i\langle B^T u,x\rangle\right)e^{i\langle u,x\rangle} \\ \int x e^{i\langle u,x\rangle}\eta_t(dx) &= -i\nabla_u\left(e^{i\langle m_t,u\rangle - \frac{1}{2}\langle u,P_t u\rangle}\right) \\ &= (m_t + iP_t u)\,\eta_t(e^{i\langle u,x\rangle}) \end{aligned}}$$

$$+ i P_t u H^T \eta_t(e^{i\langle u,x\rangle})R^{-1}(dY_t - Hm_t dt)$$

$$\begin{aligned} \mathrm{cov}_{\eta_t}(e^{i\langle u,x\rangle}, Hx) &= \\ &= (m_t + iP_t u)H^T\eta_t(e^{i\langle u,x\rangle}) - m_t H^T \eta_t(e^{i\langle u,x\rangle}) \\ &= iP_t u H^T \eta_t(e^{i\langle u,x\rangle}) \end{aligned}$$

$$= \left(-\tfrac{1}{2}\langle CC^T u,u\rangle - \langle u,BP_t u\rangle + i\langle u, Bm_t - P_t H^T R^{-1}Hm_t\rangle\right)\eta_t(e^{i\langle u,x\rangle})dt$$

(2)

$$+ i\eta_t(e^{i\langle u,x\rangle})\langle u, P_t H^T R^{-1} dY_t\rangle$$

We now make the Ansatz

$$dm_t = a_t\,dt + b_t\,dY_t$$

$$dP_t = A_t\,dt + B_t\,dY_t$$

for m_t and P_t, which is reasonable, since both, m_t and P_t are $\mathcal{Y}_{0,t}$-measurable.
Then Itô's formula implies

$$d\underbrace{e^{i\langle m_t,u\rangle - \frac{1}{2}\langle P_t u,u\rangle}}_{=:g_t(u)} = i\langle u,a_t\rangle g_t(u)\,dt + i\langle u,b_t\rangle g_t(u)\,dY_t$$
$$- \tfrac{1}{2}\langle u, b_t R b_t^T u\rangle g_t(u)\,dt$$
$$- \tfrac{1}{2}\langle u, A_t u\rangle g_t(u)\,dt - \tfrac{1}{2}\langle u, B_t u\rangle g_t(u)\,dY_t$$

(3)
$$+ \tfrac{1}{8}\langle u, B_t R B_t^T u\rangle g_t(u)\,dt - \tfrac{i}{2}\langle B_t u, R\langle u, B_t u\rangle\rangle g_t(u)^2\,dt$$

Now comparing coefficients of the three terms in dt, $\langle u,\cdot u\rangle dt$ and $i\,dY_t$ yields first that $B_t = 0$, since there is no corresponding term in (2). Hence (3) reduces to

$$dg_t(u) = i\langle u,a_t\rangle g_t(u)\,dt + i\langle u,b_t\rangle g_t(u)\,dY_t$$
$$- \tfrac{1}{2}\langle u, b_t R b_t^T u\rangle g_t(u)\,dt$$
$$- \tfrac{1}{2}\langle u, A_t u\rangle g_t(u)\,dt$$

and thus: $a_t = Bm_t - P_t H^T R^{-1} H m_t$

$$b_t = P_t H^T R^{-1}$$

$$A_t + P_t H^T R^{-1} H P_t = CC^T + BP_t + P_t B^T \quad \blacksquare$$

Algorithms: The Ensemble Kalman-Bucy filter

An important approximation of

$$E(X_t \mid Y_{0:t}) \quad \text{and} \quad Var(X_t \mid Y_{0:t})$$

in the general case of nonlinear signals and observations

$$(S) \quad dX_t = B(X_t)\,dt + C(X_t)\,dW_t$$

$$(O) \quad dY_t = H(X_t)\,dt + \Gamma\,dV_t, \quad Y_0 = 0$$

is obtained by replacing m_t (resp. P_t) in the Kalman-Bucy filtering equations by empirical means $m_t^N := \frac{1}{N}\sum_{i=1}^{N} X_t^i$ (resp. covariances $\frac{1}{N-1}\sum_{i=1}^{N}(X_t^i - m_t^N)\otimes(X_t^i - m_t^N)$), where $X_t^1, \dots, X_t^N$ solve the coupled system

$$(3) \quad \begin{aligned} dX_t^i &= B(X_t^i)\,dt + C(X_t^i)\,dW_t^i \\ &\quad + \frac{1}{N-1}\sum_{i=1}^{N} \underbrace{(X_t^i \otimes (H(X_t^i) - H^N(\mathbb{X}_t^N))^T}_{\text{empirical covariance}} R^{-1}\Big(dY_t - \frac{H(X_t^i) + H^N(\mathbb{X}_t^N)}{2}\,dt\Big) \end{aligned}$$

$$1 \le i \le N,$$

with $H^N(\mathbb{X}_t^N) = \frac{1}{N}\sum_{i=1}^{N} H(X_t^i)$.

m_t^N (resp. P_t^N) are not consistent estimates of $E(X_t \mid Y_{0:t})$ (resp. $Var(X_t \mid Y_{0:t})$) (apart from the Kalman-Bucy case), but in practice yield "good" estimation results.

Mean-field representations of the optimal filter

Another ansatz for the approximation of $E(X_t | Y_{0,t})$ (resp. η_t) is provided by constructing a stochastic process $\bar{X}_t$ with $\bar{X}_t \sim \eta_t$ for all t.

How to find $(\bar{X}_t)$?

We want to realize $(\bar{X}_t)$ as the solution of an SDE and this SDE should keep as much as possible from the structure of (S). This leads to the following ansatz:

$$
\begin{aligned}
d\bar{X}_t &= B(\bar{X}_t)\,dt + C(\bar{X}_t)\,dW_t \\
&\quad + a(t, \bar{X}_t, \bar{\eta}_t)\,dt + K(t, \bar{X}_t, \bar{\eta}_t) R^{-1} dY_t
\end{aligned}
\tag{4}
$$

for suitable coefficients a, K depending also on the distribution $\bar{\eta}_t$ of $\bar{X}_t$. In particular, $a = (a_1, \dots, a_d)$

$$K = (K_{ij})_{\substack{1 \le i \le d \\ 1 \le j \le p}}.$$

Consistency condition

Let $f \in C^2(\mathbb{R}^d)$, then Itô's formula yields:

$$
\begin{aligned}
df(\bar{X}_t) &= Lf(\bar{X}_t)\,dt + (\nabla f \cdot C)(\bar{X}_t)\,dW_t \\
&\quad + (\nabla f \cdot a)(t, \bar{X}_t, \bar{\eta}_t)\,dt + (\nabla f \cdot K)(t, \bar{X}_t, \bar{\eta}_t) R^{-1} dY_t \\
&\quad + \tfrac{1}{2} \operatorname{tr}(f'' K R^{-1} K^T)(t, \bar{X}_t, \bar{\eta}_t)\,dt
\end{aligned}
$$

Taking expectation yields:

$$d\bar{\eta}_t(f) = \bar{\eta}_t(Lf)\,dt + \bar{\eta}_t(\nabla f \cdot a(t,\cdot,\bar{\eta}_t))\,dt + \tag{5}$$
$$+ \bar{\eta}_t(\nabla f \cdot K(t,\cdot,\bar{\eta}_t))R^{-1}dY_t + \frac{1}{2}\bar{\eta}_t(\mathrm{tr}(f'' K R^{-1} K^T)(t,\cdot,\bar{\eta}_t))\,dt$$

A necessary condition that needs to be satisfied such that $\bar{\eta}_t = \eta_t$ is that (5) coincides with the KS-equation:

$$d\eta_t(f) = \eta_t(Lf)\,dt + \mathrm{cov}_{\eta_t}(f, H)R^{-1}(dY_t - \eta_t(Hf)\,dt).$$

This leads to the consistency equations:

(6a) For the dY_t-terms: $\eta_t(\nabla f \cdot K(t,\cdot,\eta_t)) \overset{!}{=} \mathrm{cov}_{\eta_t}(f, H)$

$$\Longleftrightarrow \sum_{i=1}^{d} \int \partial_{x_i} f(x) K_{ij}(t,x,\eta_t)\,\eta_t(dx) = \int f(x)\Big[H_j(x) - \int H_j(\bar{x})\,\eta_t(d\bar{x})\Big]\eta_t(dx)$$

for $j = 1, \dots, p$.

If $\eta_t(dx) = \eta_t(x)\,dx$ admits a density this yields

$$(6a)' \qquad -\sum_{i=1}^{d} \partial_{x_i}\big(\eta_t K_{ij}(t,\cdot,\eta_t)\big)(x) = \Big[H_j(x) - \int H_j(\bar{x})\,\eta_t(d\bar{x})\Big]\eta_t(x)$$

and similarly

(6b) for the dt-terms: $\eta_t\big(\nabla f \cdot a + \frac{1}{2}\mathrm{tr}(f'' K R^{-1} K^T)\big) = -\mathrm{cov}_{\eta_t}(f, H)\,\eta_t(H)$

which yields for the density

$$(6b)' \quad -\sum_{i=1}^{d} \partial_{x_i}\big(\eta_t a(t,\cdot,\eta_t)\big)(x) + \frac{1}{2}\sum_{i,j=1}^{d} \partial^2_{x_i x_j}\big(\eta_t (K R^{-1} K^T)_{ij}(t,\cdot,\eta_t)\big)(x)$$

$$= \sum_{j=1}^{p} \Big(H_j(x) - \int H_j(\bar{x})\,\eta_t(\bar{x})\,d\bar{x}\Big) \int H_j(\bar{x})\,\eta_t(\bar{x})\,d\bar{x} \quad \blacksquare$$

Equation (6a)' can be written in divergence form:

$$-\nabla(\eta_t K) = \big(H - \eta_t(H)\big)\eta_t .$$

Its solution can only be unique up to vector fields $u: \mathbb{R}^d \to \mathbb{R}^d$ that have η_t-divergence zero, i.e. $\nabla(\eta_t u) = 0$. Such vector fields correspond to η_t-measure-preserving flows.

<u>Illustration:</u> (Kalman-Bucy setting) $B(x) = Bx$, $C(x) \equiv C$, $H(x) = x$ and $\eta_t = N(m_t, P_t)$. Then (6a)' reduces to

$$-\nabla\Big(e^{-\frac{1}{2}\langle P_t^{-1}(x-m_t),(x-m_t)\rangle} K(t,x,\eta_t)\Big) = H(x-m_t)\,e^{-\frac{1}{2}\langle P_t^{-1}\cdot,\cdot\rangle}$$

$$\Longleftrightarrow \quad P_t^{-1}(x-m)\cdot K(t,x,\eta) - \nabla\cdot K(t,x,\eta_t) = H(x-m_t)$$

with one explicit solution $K(t,x) = P_t H^T$ which is exactly the "Kalman-gain" term in (2a). $\blacksquare$